Merzbacher Comparison — recipe and their true baseline

2026.07.27 — Read this before quoting any comparison number

Note crumbs so the head-to-head can be run cold. The protocol is fixed **before** we have a model, so it is not chosen after seeing our number.

what	where
split builder	`experiments/020-cachera-betaxanthin/scripts/build_merzbach`
split output	`results/merzbacher_nested_split.json`
baseline analysis	`experiments/020-cachera-betaxanthin/scripts/analyze_merzba`
baseline output	`results/merzbacher_baseline_analysis.json`
their archive (sha256-pinned)	`$DATA_ROOT/data/merzbacher2025_fcl/`

What their deposit gives us — far more than the paper implies

The manuscript reports no split and an inconsistent test N (Fig 4b 659, Table S6 649, 20 % of 811 = 162). The Zenodo **code** deposit (10.5281/zenodo.15518666 → record 15761895) has:

file	what it gives us
`data/yeast_production_test_split.csv`	the exact 640-ORF test set
`data/yeast_production_validation_split.csv`	same genes + released `label` 0/1/2 + CV `fold`
`training/yeast_training_production.py`	the exact binning rule
`figures/fig4/fig4b.csv`	per-fold accuracy, macro-F1, **MCC**, class-2 accuracy
`figures/fig4/fig4c_*.csv`	**per-gene, per-flux-sample predictions + class scores**

Only the 49 MB code archive is mirrored; `data.zip` is 23 GB of flux samples we do not need.

THEIR TRUE BASELINE — from their own shipped predictions

Aggregated to gene level by majority vote over each deletion's flux samples, exactly as their `tools/knockout_voting.py` does. Best model, `RandomForestClassifier_Resampled`:

true ↓ / pred →	low	**medium**	high
low (109)	6	**101**	2
medium (431)	2	**424**	5
high (100)	0	**82**	18

- gene-level accuracy **0.700**, majority-class rate **0.673**
- **607 of 640 genes called medium — 94.8 %**
- **18 of 100 high producers found**
- majority predictor gets 431/640; theirs gets 448/640. **The entire margin is 17 genes.**

They computed MCC and did not report it. From `fig4b.csv`:

model	accuracy	MCC	high-producer acc
RandomForest	0.698	**0.232**	0.181
RandomForest_Balanced	0.698	0.233	0.117
LogisticRegression	0.515	**0.041**	0.233
LinearSVC_Resampled	0.445	**0.031**	0.273

MCC 0.23 is weak but **not zero** — there is real signal, concentrated in the majority class. Be fair in any writeup:

they claim “promising accuracy”, not a significant gain.

The decisive structural point: every model that finds more high producers has *worse* overall accuracy, because the only way to call more highs is to stop calling everything medium. **Accuracy actively selects against the capability the task is about.**

The recipe, fixed in advance

1. **Ground truth = THEIR RELEASED LABELS**, never labels we re-derive. Their rule min-max scales to [0,1] then cuts at 0.40 / 0.65, so the cuts depend on the observed extremes. Applied to our (larger) copy of the same screen it gives 107/476/56 against their 109/431/100 — 81.2 % agreement, and **we call barely half as many high producers**. Same rule, same screen, different classes. Re-deriving would compare different tasks.
2. **Bin OUR predictions with thresholds fitted on the TRAIN split only.** Fitting on test leaks the class distribution, which on a 67 %-majority problem is most of the answer.
3. **Primary metrics are imbalance-immune**: Spearman, and top-*k* enrichment on high producers. Report **MCC** next to accuracy — they had it, so we should show it.
4. **Also report accuracy + per-class + high-producer recall**, so the comparison lands on their terms too.
5. **State the fraction of genes we call medium.** If ours is also ~95 %, we reproduced their failure mode rather than beat it, and must say so.

Reconciliations to carry into any writeup — report, never resolve silently

- **639 of 640** of their test genes are in our test split. The one missing is `YBR011C` = **IPP1**, which is **essential** (SGD S000000215, null inviable). No haploid deletion collection contains *ipp1*Δ, so no screen could have measured it — **their test set contains a gene with no possible deletion phenotype.**
- **`PPA1` is an alias of both IPP1 and VMA16.** The screen has ONE `PPA1` row; their split has BOTH YBR011C and YHR026W as separate test genes. At least one of their test entries plausibly carries **VMA16’s** value scored as if it were IPP1. Our screen-local disambiguation (`PPA1`→VMA16, `FEN1`→ELO2) is documented in the split builder, with the reasoning recorded so it can be re-derived if its premises change.
- **640 vs the paper’s 811** — a 171-gene gap the Methods do not count (“sampling failed to converge”).
- **Their validation folds appear drawn from the same 640 genes as their test list** (`yeast_training_production.py:` which would mean model selection touched test data. FLAGGED, not claimed — confirming needs `yeast_single_knockouts.npz` from the 23 GB archive.
- Our Cachera build is **stale** w.r.t. the name resolver (issue #195). The split is built from the RAW screen + resolver so it does not inherit that, but **training data still will** until a rebuild.

What we can report that they cannot

- **Regression at all.** They tried and abandoned it — *“challenging with the limited number of knockouts at the high and low ends.”* Our betaxanthin noise ceiling is **r = 0.914** (reliability 0.836, median 15 replicates), so regression is available to us.
- **The ~3,900 non-metabolic deletions.** FCL has no flux cone for a gene outside Yeast9, so it cannot make a prediction at all. We predict all ~4,700. That is a category difference, not a tie-break.

Where our model stood when the sweeps were paused

`020-cachera-betaxanthin`, study `betaxanthin_002`, 33/40 trials before cancellation: **val Pearson 0.430** — 47 % of the 0.914 ceiling — with `prot_T5_all` / learnable **False** / `crps` / L=2 / hidden 90 / 6.5e-5 / decayed / yeo-johnson. The top-5 configs agreed closely, so it is a plateau rather than a lucky trial. Companion arms: `021-ozaydin-beta-carotene` (Spearman 0.223) and `022-mulleder-metabolome` (0.180, up from a historical $\leq$ 0.025).

Related: [[plan.cgt-metabolism-flux-layer.2026.07.26]] · [[plan.simb-2026-multimodal-cgt.2026.07.21]]

2026.08.02 - CGT vs Flux Cone Learning on the Cachera betaxanthin screen

Naming, because the two get conflated. *Cachera* is the genome-wide betaxanthin DELETION SCREEN – the data, ~4,700 single deletions, and the ground truth for both models. *Flux Cone Learning* (FCL) is Merzbacher et al.’s METHOD, which learns from flux samples drawn through a genome-scale metabolic model;

`RandomForest_Resampled` is its best variant and is what "FCL RF" means throughout. The comparison is **CGT vs FCL, both scored on the Cachera screen** – not "us vs Cachera".

All figures regenerate from `experiments/020-cachera-betaxanthin/scripts/plot_merzbacher_comparison.py`, built from the 10 finished Delta `020_v4` cells (`$DATA_ROOT/test-predictions/`) and the FCL paper's shipped `figures/fig4/fig4c_*.csv`.

- **639 genes** carry their label, a CGT prediction and a raw screen value. Their 640th (IPP1/YBR011C) is essential – no deletion strain exists, so no screen could measure it.
- **FCL model = RandomForest_Resampled**, the best of their four (the others read 0.56-0.64 gene-level against RF's 0.700).
- **CGT = ONE cell, selected by VALIDATION** (`s09_L6_maskon_lr0.0001_yj`, val 0.3639), never by its score on their test genes.
- **Provenance asserted:** our re-derivation of their deployed gene-level vote reproduces their published accuracy (0.7011 vs 0.700) or the script exits.

TL;DR

1. **The published accuracy comparison is empty.** 0.701 vs a 0.674 majority rate, achieved by calling 95 % of genes medium. Rank-matched, both models fall to ~0.55. (Fig 2)
2. **Both models DO have real signal, concentrated at the top of the ranking.** ~5x enrichment in the top 10-25 genes, $p < 1e\text{-}06$ – and essentially nothing across the bulk. (Fig 3, 4)
3. **CGT and their RF are roughly tied where both can be measured**, and disagree about which genes (rank correlation -0.128). (Fig 1, 3, 5)
4. **The real difference is COVERAGE, not accuracy.** Their features are flux samples from yeast-GEM, so their method has no input at all for a deletion outside the metabolic model; their own test set is 99.8 % inside it. CGT scores those genes and finds high producers there at 4.4x enrichment ($p = 2e\text{-}05$). (Fig 8)

Fig 1 – per-gene spread (639 points)

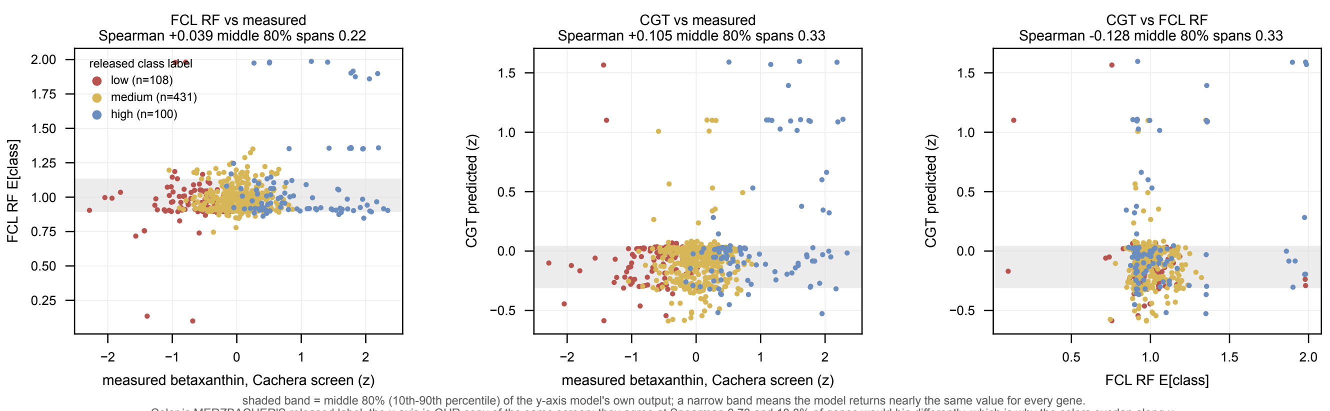

Fig 1. Both models emit a nearly constant value; the two disagree about which genes. 80 % of their genes fall inside an E[class] band of 0.22 on a 0-2 scale, 80 % of CGT's inside 0.33 z-units. Colour is the FCL paper's released class label, x is OUR copy of the same screen.

pair	Pearson	Spearman
measured vs FCL RF	+0.244	**+0.039**
measured vs CGT	+0.294	**+0.105**
FCL RF vs CGT	+0.108	**-0.128**

Pearson exceeds Spearman on both model rows because each gets a few extreme genes right and is otherwise flat. In RANK the two models are slightly ANTI-correlated.

Fig 2 – why the accuracy comparison is degenerate

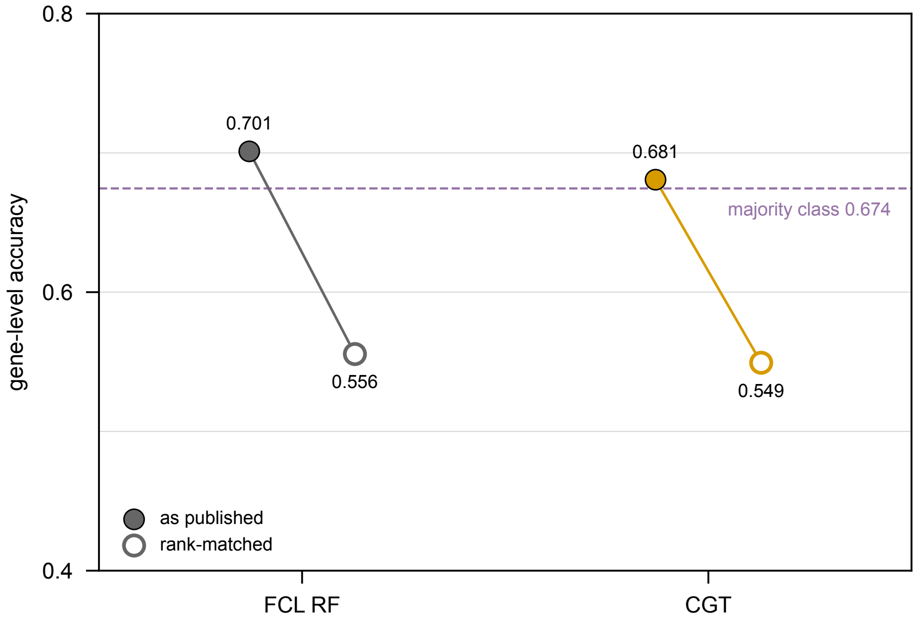

Fig 2. Accuracy above the majority line is the conservative-majority strategy, not discrimination – for both models. RF's 0.701 clears the 0.674 majority rate by 0.027 by calling 95 % of genes medium; CGT's absolute binning sits at 0.681 doing the same. Forced to the true class marginal, RF drops to 0.556 and CGT to 0.549. Evidence, not a result.

Fig 3 – precision@k for high producers

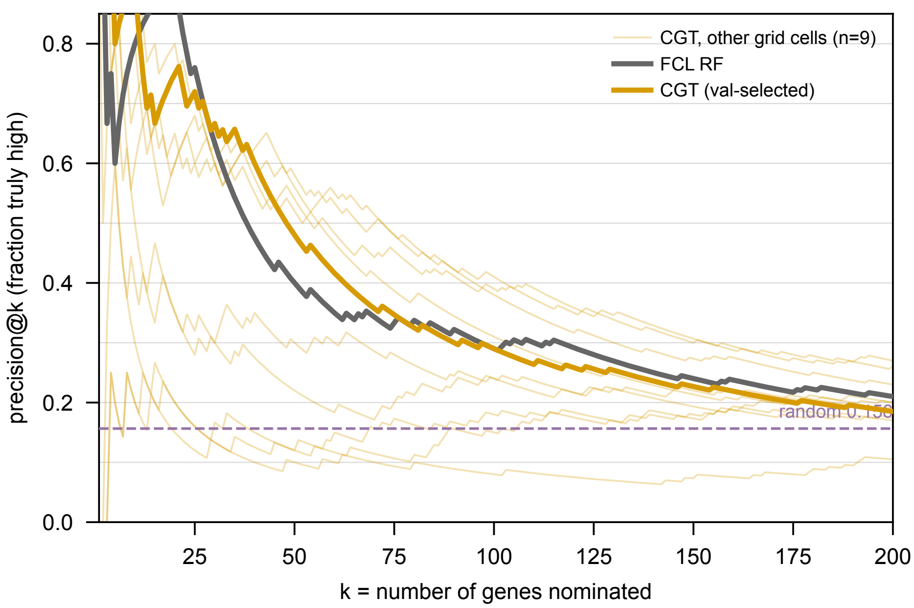

Fig 3. Both models are strongly enriched at the top of their ranking. Of the top k genes each nominates, the fraction truly high; random is 0.156.

model	k=10	k=25	k=50
FCL RF	8 (5.1x, p=1e-05)	19 (4.9x, p=8e-12)	20 (2.6x, p=1e-05)
CGT	**9** (5.8x, p=4e-07)	18 (4.6x, p=1e-10)	**24** (3.1x, p=2e-08)

Nine of CGT's top ten are genuine high producers. The two models track each other closely and both approach the base rate by k=150. The nine unselected CGT cells span 0.10-0.65 at k=50 – cell choice matters more than the CGT-vs-RF gap.

Fig 4 – score by class, as percentile rank

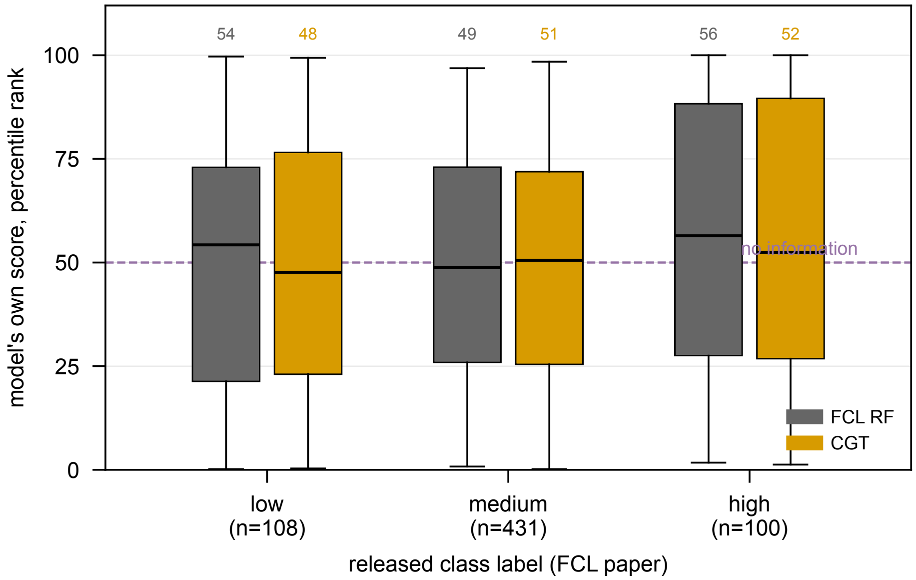

Fig 4. Neither model orders the classes across the bulk of the population. Percentile rank puts two models with incomparable units on one axis; a separating model shows three stepping boxes.

model	low	medium	high
FCL RF	54	49	56
CGT	48	51	52

Read this WITH Fig 3, not instead of it. A median is a bulk statistic and the top 25 genes are 4 % of the population, so it barely moves. Together the two figures say the signal is real and confined to the tail – not that there is none.

Fig 5 – ROC for high-producer detection

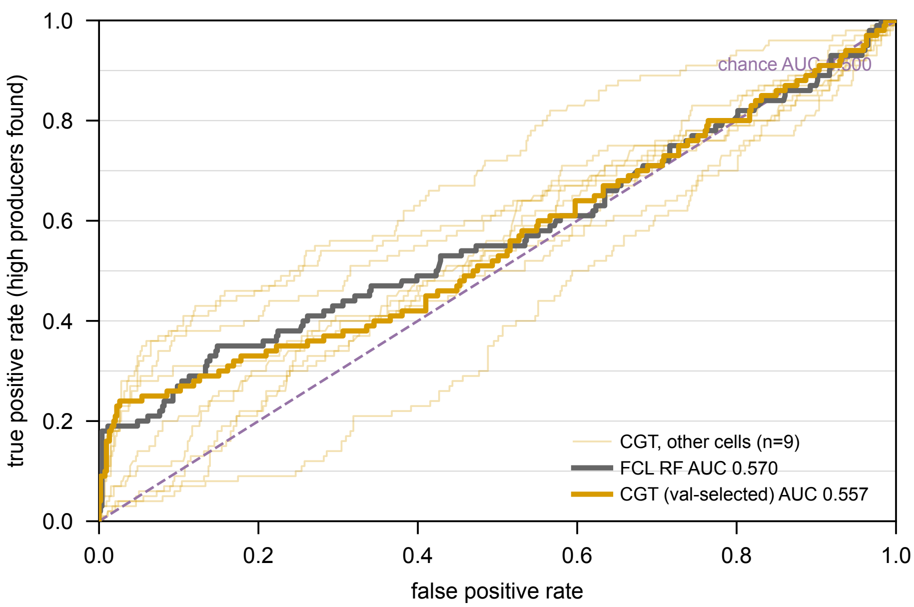

Fig 5. Threshold-free, and it goes marginally to their RF: AUC 0.570 vs CGT's 0.557. Both near the 0.500 chance line, because AUC integrates over the whole ranking where neither model has much. The honest summary is metric-dependent: CGT wins global Spearman, RF wins AUC, precision@k is a wash, and no gap is large.

Fig 6 – how much is cell selection?

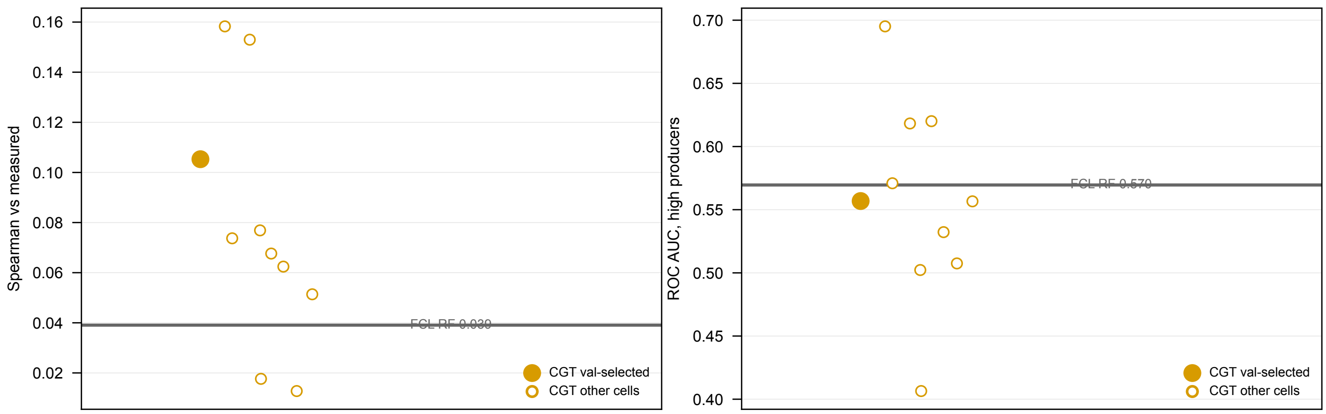

Fig 6. The spread across grid cells is wider than the gap between models. CGT cells span 0.013-0.158 Spearman (8 of 10 beat RF's 0.039) and 0.406-0.695 AUC (only 3 of 10 beat RF's 0.570, and the val-selected cell is not among them). Four low points are the lr=1e-3 cells that collapsed to a constant predictor. Any single-number claim is dominated by which cell is quoted, which is why the cell is fixed by validation in advance.

Fig 7 – do their labels track our copy of the screen?

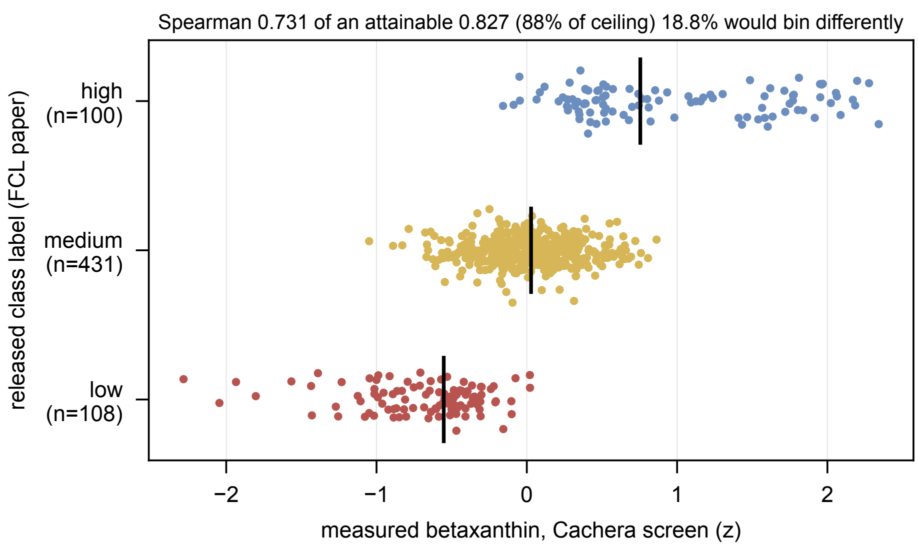

Fig 7. Their labels reach 88 % of the attainable ceiling – a modest caveat, not a problem. Spearman 0.731 against a ceiling of 0.827 (a 3-level ordinal cannot reach 1.0 against a continuous variable because of its own ties, so the raw number is uninterpretable alone). The residual is real: **18.8 % of genes would bin differently** under a pure rank-split of our values, so a model perfectly predicting OUR measurement would still be scored wrong on ~19 %.

Fig 8 – the capability gap: metabolic vs non-metabolic genes

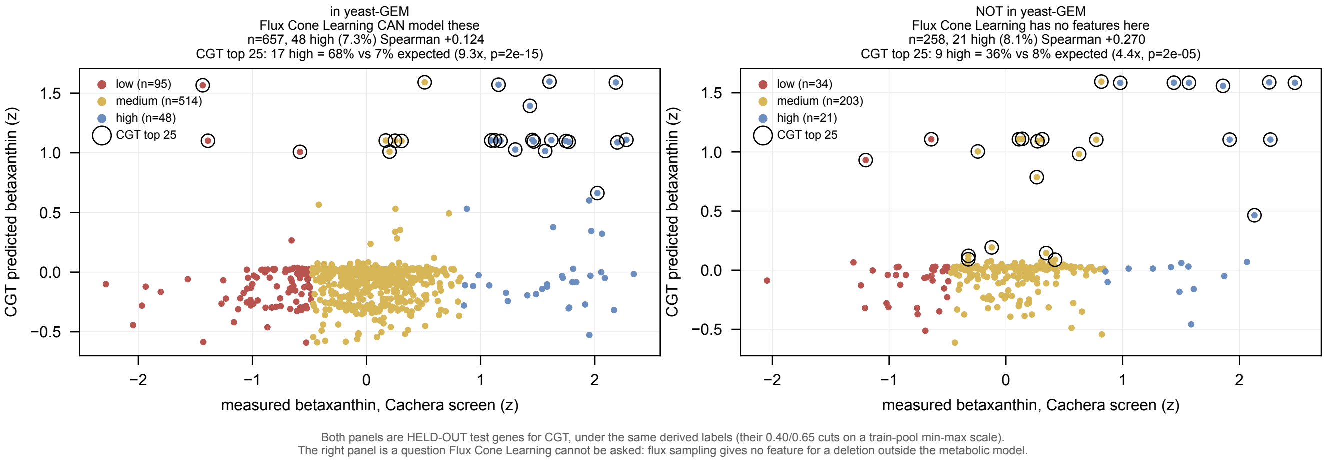

Fig 8. CGT predicts high betaxanthin producers among deletions the flux-based method cannot represent. Held-out test genes split by yeast-GEM membership, identical derived labels on both sides (their 0.40/0.65 cuts on a train-pool min-max scale). Circled = CGT's top 25.

	in yeast-GEM	NOT in yeast-GEM
genes	657	258
true high (derived)	48 (7.3%)	21 (8.1%)
CGT Spearman vs measured	+0.124	**+0.270**
CGT top 25: high found	17 (68%)	9 (36%)
enrichment over base rate	9.3x, p=2e-15	**4.4x, p=2e-05**
Flux Cone Learning prediction	available	**none possible**

the FCL paper's test set is **639/640 (99.8 %) inside yeast-GEM**; of our other test genes only **21/294 (7.1 %)** are. That is what the constraint looks like in practice: flux sampling yields no feature for a deletion the metabolic model does not contain, so FCL's output for those genes is not poor, it is undefined.

CGT's rank correlation is *higher* on the non-metabolic set (+0.270 vs +0.124). Both enrichments are strongly significant.

Caveats, and they matter for how hard this can be pushed. (i) Labels for the non-GEM genes are DERIVED by us – the FCL paper never labelled them – so both panels use our derivation and the comparison to make is panel-vs-panel, not panel-vs-their-paper. (ii) n=258 with 21 positives is modest; the interval on 4.4x is wide. (iii) "FCL cannot model these" is a STRUCTURAL claim about flux-sample features, not an experiment we ran on their code.

Method notes

Can their side be ranked? Yes – they release more than classes. `fig4c_*.csv` carries `score0/score1/score2` (P(low)/P(med)/P(high), summing to 1.0) for 124 flux samples per gene across 640 genes, aggregating to 558 distinct gene-level values via `E[class] = p_med + 2*p_high`. Two checks: TIES ARE INERT (82 genes share a value, but over 200 random permutations accuracy moves 0.5563 +- 0.0008 and high recall not at all); and THE AGGREGATION IS NOT LOAD-BEARING, the one used being the most generous to them – `E[class]` 0.5556/0.290, `p_high - p_low` identical, `p_high` alone 0.5446/0.280.

The rank-matched numbers use an ORACLE class marginal. Counts come from the TRUE TEST labels (108/431/100), so both models are handed the marginal of the answer. The sibling `evaluate_merzbacher_head_to_head.py` uses the TRAIN-POOL marginal (107/504/28) and is leak-free. Same marginal on both sides, so the RELATIVE claim holds; the ABSOLUTE values are not deployable accuracy:

marginal	RF acc	CGT acc	RF hi-rec	CGT hi-rec
test counts (oracle, used here)	0.5556	0.5493	0.290	0.290
train-pool counts (leak-free)	0.6228	0.6166	0.190	0.190

RF leads by ~0.006 either way, high recall identical either way. The pool expects 28 high producers where the test set holds 100: their test genes are enriched for extremes.

Diagnostics not carried as figures (regenerate with `--all`): class distribution, per-class recall, rank-matched confusion matrices. Rank-matched, CGT reaches 0.29 recall on high producers vs their published 0.18 and 0.19 on low vs 0.06, paid for in medium recall (0.70 vs 0.98); the rank-binned labels agree on only 52 % of genes.

Caveat on MCC. The 0.205 quoted elsewhere is their PER-FLUX-SAMPLE MCC; they never published a gene-level MCC. Accuracy and high-producer recall are gene-level on both sides.

Status. 10 of 24 Delta `020_v4` cells finished; wave C (L=4) just started. Re-running the rsync and the scripts picks up the rest, and the CGT points can move.

Slide set: Fig 8 is the argument (coverage, not accuracy). Fig 2 + Fig 3 set it up – the published metric is empty, but both models do find high producers at the top. Fig 4 and Fig 6 are the qualifiers.